



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
20-05-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [SCORE-II (STAR BATCH)]

PAPER-1

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

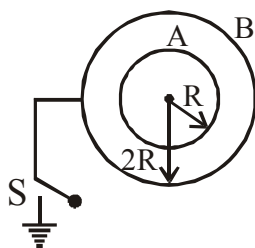
SECTION-I(i) (Maximum Marks: 28)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If electron of hydrogen atom is replaced by another particle of same charge but of double mass then
 (A) Bohr's radius will increase
 (B) Ionisation energy of atom will be doubled
 (C) Speed of new particle in given state will be lesser than electrons speed in same orbit
 (D) Gap between energy levels will now be doubled
2. A metal spherical shell of mass m is filled **completely** with nonviscous water of mass $m/2$ and is rolled down an inclined plane of height h . The total kinetic energy of shell plus water and the velocity of the centre of the shell when it reaches the bottom of the plane is K and v respectively. If water in the shell is freezed and then it is rolled down, the corresponding values of kinetic energy and velocity are K' and v' respectively. Then :
 (A) $K = mgh$ (B) $K' = mgh$ (C) $v = v'$ (D) $v > v'$

Space for Rough Work

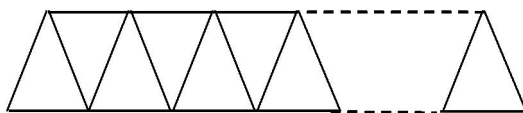
3. A spherical planet of radius R has spherically symmetrical distribution of mass density, varying as square of the distance from the centre, from zero at centre to maximum value ρ_0 at its surface.
- (A) The value of escape velocity of a mass m at the surface of planet is $\sqrt{\frac{4\pi G \rho_0 R^2}{5}}$.
- (B) The value of acceleration due to gravity 'g' varies inside the planet as cube of the distance from centre.
- (C) The value of escape velocity from surface is same as the escape velocity from another planet of same total mass & radius but having uniform mass density.
- (D) The energy required to impart escape velocity to particles of masses ' m ' & ' $2m$ ', at the surface of planet, will be in ratio 1:2.
4. Two concentric conducting spherical shells of radii R and $2R$ having charges q_A and q_B and potential $2v$ and $3v/2$ respectively. Now shell B is earthed by closing switch S . Let charges on spherical shells A & B become q'_A & q'_B respectively, then



- (A) $q_A/q_B = \frac{1}{2}$ (magnitude).
- (B) $q'_A/q'_B = 1$ (magnitude).
- (C) potential of A after earthing becomes $\frac{3v}{2}$.
- (D) potential difference between A and B ($v_A - v_B$) after earthing becomes $\frac{v}{2}$.

Space for Rough Work

5. The magnetic flux (ϕ) linked with a coil depends on time t as $\phi = at^n$, where a is a constant. The induced e.m.f. in the coil is e :
- (A) If $0 < n < 1$, $e = 0$
 (B) If $0 < n < 1$, $e \neq 0$ and $|e|$ decreases with time
 (C) If $n = 1$, e is constant
 (D) If $n > 1$, $|e|$ increases with time
6. Sound waves from different source are interfering and pressure at some point in space is given as –
 $P = P_0 \sin (200 \pi t + \pi/2) \cos (3 \pi t + \pi/3)$
 (A) The frequency of one source is above 100Hz and other source is below 100Hz.
 (B) Beat frequency is 3 Hz
 (C) Beat frequency is 200 Hz
 (D) Beat frequency can be 3 Hz as well as 200 Hz
7. n number of identical equilateral prism are kept in contact as shown in figure. If deviation through a single prism is δ . Then - (n, m are integers)



- (A) If $n = 2m$, deviation through n prisms is zero.
 (B) If $n = 2m + 1$, deviation through system of n prisms is δ .
 (C) If $n = 2m$, deviation through system of n prisms is δ .
 (D) If $n = 2m + 1$, deviation through system of n prisms is zero.

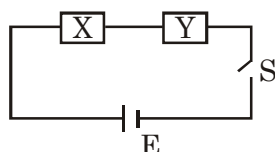
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

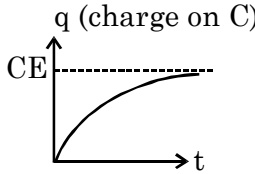
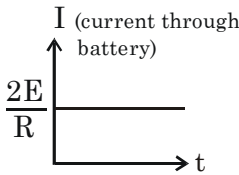
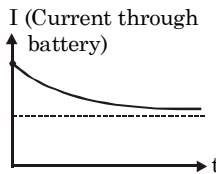
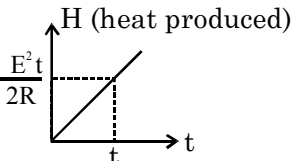
- This section contains **SIX** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

Following is a circuit containing an ideal battery of emf E in which the elements X & Y are unknown & may be a capacitor C , a resistor R , a battery of emf E , or may be their combination as mentioned in each case. The connecting wires are perfectly conducting.



The switch S is initially open & it is closed at $t = 0$.

Column-1	Column-2	Column-3
(I) Element X is R Element Y is C	(i) Entire energy (non-zero) supplied by batteries is dissipated as heat	(P) 
(II) Both elements X & Y are R each	(ii) Total charge drawn from the battery(s) after long time is $\frac{CE}{2}$	(Q) 
(III) Element X is R , element Y is a parallel combination of R and C	(iii) Total heat produced in circuit is less than the energy supplied by the battery(s)	(R) 
(IV) Element X is R element Y is battery E joined with same polarity	(iv) $I = 0$ just after S is closed as well as after long time	(S) 

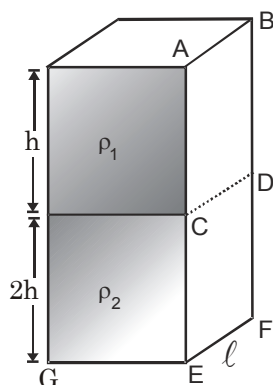
Space for Rough Work

8. Of the given 4 situations, which one has maximum value of current just after closing the switch ?
(A) I, iii, P (B) IV, i, Q (C) II, iii, P (D) III, i, S
9. In which case the charge drawn from the battery(s) during $t = 0$ to $t = RC$ is the least ?
(A) II, i, S (B) I, iii, P (C) III, ii, Q (D) IV, i, Q
10. In which case the current through battery(s) has different non zero values just after closing the switch & after a long time ?
(A) II, i, S (B) I, iv, Q (C) IV, i, Q (D) III, iii, R

Space for Rough Work

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given the three columns of the following table.

A cuboid is filled with liquid of density ρ_2 upto height $2h$ & with liquid of density ρ_1 , upto height h as shown in the figure and assume pressure at top to be zero.



Column-1	Column-2	Column-3
(I) Force on face ABCD due to liquid of density ρ_1	(i) zero	(P) Effectively acts at centre of that face
(II) Force on face ABCD due to liquid of density ρ_2	(ii) $\frac{\rho_1 g h^2 \ell}{2}$	(Q) Effectively acts at $\frac{h}{3}$ above the bottom of that face
(III) Magnitude of increase in force on face CDEF due to filling liquid of density ρ_1	(iii) $2\rho_1 g h^2 \ell$	(R) Effectively acts of at $\frac{4h}{3}$ below the top of that face
(IV) Force on face CDEF if liquid of density ρ_2 only is present	(iv) $2\rho_2 g h^2 \ell$	(S) Cannot produce a net torque about bottom edge of that face
11. Which force has the second highest magnitude (given $\rho_1 < \rho_2$) ?		
(A) I,ii, Q	(B) IV, i,Q	(C) II,i, S
(D) III,iii, P		
12. Which force produces minimum (Non zero) torque about side E F (given $\rho_1 = \rho_2$) ?		
(A) I,ii, Q	(B) II,i, S	(C) III,iii, P
(D) IV, iii, S		
13. Which force is equal to the total weight of liquid ρ_2 [Given GE = h]?		
(A) I, iii, P	(B) II,i, S	(C) III,iii, P
(D) IV, iv, R		

Space for Rough Work

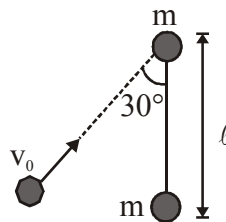
SECTION-II : Numerical Value Type (Up to second decimal place)

No question will be asked in section II

SECTION-III : (Maximum Marks : 15)

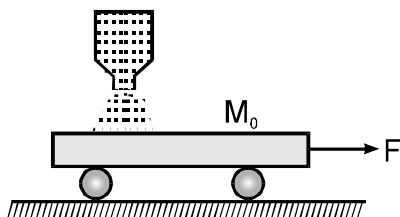
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Zero Marks : 0 In all other cases.

1. A small putty of mass $m = 1$ kg and speed $v_0 = 4$ m/s, strikes a dumbbell placed on a smooth table as shown in figure. Putty is moving in the plane of the dumbbell. The putty sticks to dumbbell ball which is to be treated as particle. How much thermal energy is produced (in J). Consider rod of dumbbell to be massless.

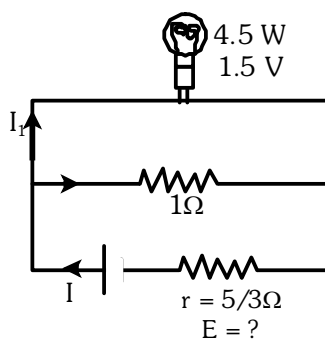


Space for Rough Work

2. In the figure shown a cart moves on a smooth horizontal surface with a constant velocity v acted upon by an external constant force of magnitude F . Sand falls on to the cart with negligible velocity at constant rate μ kg/s and sticks to the cart. Find N if the power supplied by external agent applying the force F is μv^N .

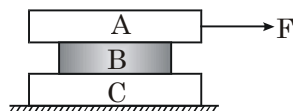


3. A torch bulb rated at 4.5 W, 1.5 V is connected as shown in the figure. To make the bulb glow at full intensity, what should be the emf (in volt) of the cell?

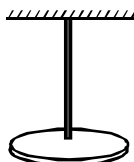


Space for Rough Work

4. Given $m_A = 30 \text{ kg}$, $m_B = 10 \text{ kg}$, $m_C = 20 \text{ kg}$. Between A and B; $\mu_1 = 0.3$, between B and C; $\mu_2 = 0.2$ and between C and ground $\mu_3 = 0.1$. The least horizontal force which can start the motion of any part of the system of three blocks resting upon one another as shown in figure is F . Write the value of $F/12$ ($g = 10 \text{ m/s}^2$)



5. A circular disc suspended from a torsional wire, oscillates with a time period T when twisted by a small angle θ from equilibrium and released.



If another disc of same material but twice the radius and one fourth thickness is made to oscillate (alone) with same wire, its time period is found to be nT . Write the value of n .

Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) (Maximum Marks: 28)

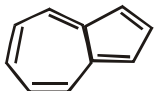
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1. Which of the following statement(s) is/are correct ?

- (A) Glucose & Fructose are optical isomers.
 (B) Fructose can be oxidised by tollens reagent.
 (C) Monosaccharides containing six membered ring can be called pyranose sugar.
 (D) Sucrose is a disaccharide and it contains two glycosidic linkages.

2. Which of the following statement(s) is/are incorrect ?

- (A) Bakelite is a polyamide
 (B) All Biodegradable polymers are natural polymers

(C) Azulene  has zero dipole moment.

(D) Sulphonation of C_6H_6 , C_6D_6 & C_6T_6 by Electrophilic aromatic substitution mechanism proceed at same rate.

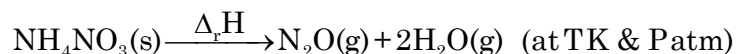
Space for Rough Work

3. Select **INCORRECT** statement(s) :

- (A) pK_a value of $B(OH)_3(aq.)$ increases on adding glycerol.
- (B) aqueous solution of $Ca(OH)_2$ can not be used for removal of temporary hardness of water.
- (C) Crystals of borax on heating does not produce any neutral oxide.
- (D) Two layers of solid orthoboric acid are connected together by hydrogen bonds.

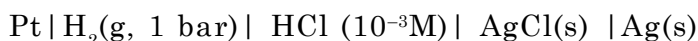
4. For thermal decomposition of $NH_4NO_3(s)$:

The correct information(s) is/are -



- (A) Heat of reaction is positive
- (B) Entropy change of the reaction must be $\frac{\Delta_r H}{T}$
- (C) Heat of reaction at constant volume is equal to $(\Delta_r H - 3 RT)$
- (D) $\Delta_r G$ of this reaction may be negative at high temperature.

5. Select the correct statement(s) regarding following galvanic cell taken at temperature TK.



Given that : $\frac{2.303RT}{F} = 0.06 \text{ volt}$

& $E^\circ_{Ag|AgCl|Cl^-} = -0.18 \text{ volt}$

- (A) Cell potential of this cell is 0.54 volt.
- (B) During discharging, $H_2(g)$ converts into $H^+(aq.)$
- (C) $H_2(g)$ is obtained at cathode during working of this galvanic cell.
- (D) $E^\circ_{Ag^+|Ag}$ is equal to 0.78 volt if K_{sp} of $AgCl$ is 1×10^{-10} .

Space for Rough Work

6. Select the **CORRECT** option(s) about Valence Bond Theory regarding co-ordination compounds:
- (A) It does not explain visible spectra of complexes.
 - (B) It is not able to explain diamagnetic behaviour of $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ even Co^{3+} is paramagnetic.
 - (C) It is unable to differentiate between weak and strong field ligands.
 - (D) It makes exact predictions regarding the tetrahedral and square planar structures of 4-coordinate complexes.
7. Which of the following step(s) is/are involved in extraction of iron from haematite ore ?
- (A) Gravity separation
 - (B) Reduction by CO
 - (C) Self-reduction
 - (D) Slag formation

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **SIX** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
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Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

For given solution in the questions molarity can be treated as molality and solute(s) is/are nonvolatile in nature.

Given that : K_a of $\text{CH}_3\text{COOH} = 10^{-5}$; K_f = Molal depression constant
 K_a of $\text{HCOOH} = 10^{-4}$; K_b = Molal elevation constant

Column - I (Colligative properties)	Column - II (One of the product formed during electrolysis at Pt - electrode)	Column - III (Nature of aqueous solution at 25°C)
(I) $\Delta T_b \approx 0.3 K_b$	(A) Metal becomes deposited at cathode	(P) pH = 8.5
(II) $\pi \approx 0.4 RT$	(B) H_2 gas at cathode	(Q) pH = 4.7
(III) $\Delta T_f \approx 0.2 K_f$	(C) H_2 gas at anode	(R) pH ≈ 7
(IV) $\pi \approx 0.3 RT$	(D) O_2 gas at anode	(S) Weakly basic solution
8. Which of the following option is correctly matched for 0.1 M HCOOK(aq.)		
(A) (I),(B),(S)	(B) (IV), (B), (S)	(C) (III),(D), (Q) (D) (III),(C),(P)
9. Select incorrect match for 0.1 m $\text{Ca(NO}_3)_2$ aqueous solution.		
(A) (I),(B),(R)	(B) (III),(A),(P)	(C) (IV),(D),(R) (D) (I),(D),(R)
10. Select the correct match for aq. solution having 0.1 M CH_3COONa and 0.2 M CH_3COOH .		
(A) (II),(B),(Q)	(B) (II),(A),(Q)	(C) (IV),(A),(S) (D) (I),(C),(R)

Space for Rough Work

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given the three columns of the following table.

Column-I (Name reaction or test)	Column-II (Related Reagent or intermediate)	Column-III (miscellaneous)
(I) Hoffman bromamide reaction	(i) Carbene	(P) Reaction involves upgradation of chain
(II) Reimer tieman reaction	(ii) CHCl_3	(Q) Reaction involves chain degradation
(III) Aldol condensation	(iii) KOH	(R) Synthesis of amine takes place
(IV) Isocyanide test	(iv) Br_2	(S) Used to test primary amines

11. Which of the following option is correctly matched ?

- (A) (I), (iv), (P) (B) (II), (iv), (Q) (C) (III), (i), (R) (D) (IV), (ii), (S)

12. Select incorrect match :

- (A) (I), (iv), (Q) (B) (II), (i), (R) (C) (III), (iii), (P) (D) (IV), (iii), (S)

13. How many reaction/test from column-I can use KOH as reagent ?

- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)

No question will be asked in section II

SECTION-III : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
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1. Lime stone(X) $\xrightarrow{\Delta}$ Solid (A) + gas (B)

Other Binary Compound of Calcium(Y) $\xrightarrow{\text{H}_2\text{O}}$ Solution (C) + Gas (D)

Gas (B) turns solution (C) milky but does not react with KMnO_4/H^+ .

Gas (D) produce white ppt. in tollen reagent and it also produce benzene when passed through red hot iron tube.

The difference of oxidation state of same central atom in (B) & (D) is :

2. How many comparison(s) are **INCORRECT** among given ?

(i) $\text{H} > \text{Li}$ (ionisation energy)

(ii) $\text{Li} > \text{Be}$ (size)

(iii) $\text{Na} > \text{Rb}$ (size)

(iv) $\text{O} > \text{N}$ (size)

(v) $\text{S} > \text{P}$ (ionisation energy)

(vi) $\text{C} > \text{O}$ (size)

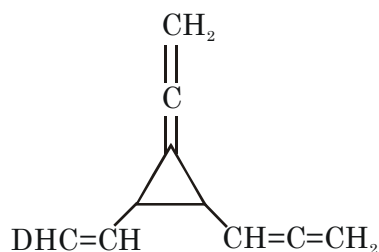
(vii) $\text{B} > \text{Al}$ (size)

(viii) $\text{F} > \text{Cl}$ (ionisation energy)

(ix) $\text{N} > \text{O}$ (ionisation energy)

Space for Rough Work

3. Number of non-radioactive alkali metals forming superoxide as major product on heating with excess $O_2 = x$,
 Number of II-A metals of periodic table whose hydrated halides suffer hydrolysis on heating = y .
 Number of non radio active alkali metals which dissolve in liquid ammonia and produce blue colour solution = u
 The value of $(x + y - u)$ would be....
4. Total number of stereo isomers corresponding to structure



Space for Rough Work

5. The formation of dextrorotatory isomer A from its levorotatory isomer B in water is reversible reaction having 1st order kinetics in both direction. Racemic mixture is the equilibrium product if reaction starts from either A or B. Analysis of polarimeter experiment showed that 10% of B was converted into A in 230.3 sec at 25°C.

($\log 5 = 0.7$, $\log 3 = 0.48$ and $\log 2 = 0.3$)

Based on given information select number of correct statement(s) among following.

- (A) Rate constant of forward reaction at 25°C is $5 \times 10^{-4} \text{ s}^{-1}$
 (B) Rate constant of backward reaction at 25°C is $5 \times 10^{-4} \text{ s}^{-1}$
 (C) Activation energy of forward reaction is equal to activation energy of backward reaction.
 (D) Equilibrium constant (K) at 25°C of $B \rightleftharpoons A$ is 1.
 (E) Equilibrium constant increases by increasing temperature.
 (F) Molar entropy of A and B at 25°C must be same.
 (G) $\Delta_r H$ of the process is zero.
 (H) $\Delta_r G^\circ$ of the process is zero.
 (I) $\Delta_r G$ of the process $B \xrightleftharpoons[k_b]{k_f} A$ is zero.

(l-form) (d-form)

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) (Maximum Marks: 28)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $\vec{a}, \vec{b}, \vec{c}$ be vectors such that $\vec{b} \cdot \vec{c} = 2\vec{a} \cdot \vec{a} = 3\vec{b} \cdot \vec{b} = \vec{a} \cdot \vec{b} \times \vec{c} = 1$ and $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0$. Now if $\vec{a} \times \vec{b}$ is expressed as $x\vec{a} + y\vec{b} + z\vec{c}$, then
 (A) $x < 0$ (B) $y < 0$ (C) $z < 0$ (D) $x + y + z < 0$
2. Let f be a twice-differentiable real valued function satisfying the equation :
 $x + f(x)f'(x) + f'(x)f''(x) = 0$, then
 (A) $\lim_{x \rightarrow 0} xf(x) = -1$ (B) $\lim_{x \rightarrow 0} \frac{f'(x)}{\ln|x|} = 0$
 (C) $\lim_{x \rightarrow 0} xf(x)f'(x) = 1$ (D) $\lim_{x \rightarrow 0} \frac{f(x)f'(x)}{\ln|x|} = 0$

Space for Rough Work

3. Let the conics $x^2 = xy + 1$ and $xy = x^2 + 1$ have eccentricities e_1 and e_2 respectively then
 (A) $e_1 > e_2$ (B) $e_1 < e_2$ (C) $e_1 \cdot e_2 = 2$ (D) $e_1^{-2} + e_2^{-2} = 1$
4. Let $A = [a_{ij}]_{3 \times 3}$ be a matrix where $a_{ij} \in \{0, 1\} \forall 1 \leq i, j \leq 3$. Let $s_i = \sum_{k=1}^3 a_{ik}$ and $t_j = \sum_{k=1}^3 a_{kj}$ then the probability that the product $(s_1 s_2 s_3 t_1 t_2 t_3)$ is an odd integer, is
 (A) $\frac{1}{2}$ (B) $\frac{1}{8}$ (C) $\frac{1}{16}$ (D) $\frac{1}{32}$
5. If A is an orthogonal matrix of order 3×3 and $\det(A) = 1$ then which of the following matrices is/are **NOT** invertible
 (A) $A - I$ (B) $A^2 - I$ (C) $(A - A^T)^2$ (D) $A - (\det(A))I$
6. Let for $n \in \mathbb{N}$; $(2x^2 - 2x + 1)^n = \sum_{r=0}^{2n} (-1)^r a_r x^r$ then
 (A) $\sum_{r=0}^{2n} a_r = 1$ (B) $\sum_{r=1}^{2n} (-1)^r a_r = 0$
 (C) $\sum_{k \geq 0} {}^n C_k {}^{2n-2k} C_r = 2^{n-r} a_r$ (D) $\sum_{r=0}^{2n} ((-1)^r a_r a_{2n-r}) = 0$ if $n = \text{odd}$
7. Let $(x - k)^2 + y^2 = r^2$, ($k, r > 0$) is the largest circle, no part of which is lying outside the ellipse $3x^2 + 4y^2 = 12$. Now
 (A) if $k < \frac{1}{2}$ then $r + k = 2$ (B) if $k > \frac{1}{2}$ then $r + k = 2$
 (C) if $k < \frac{1}{2}$ then $r + k < 2$ (D) if $k > \frac{1}{2}$ then $r + k < 2$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **SIX** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

Consider the following path traced by point (x,y) on the curve $x^a + y^a = 1$ lying on the xy plane.

Column-I (Condition)	Column-II (Nature of path)	Column-III (A tangent to the path)
(I) $a = -1$	(i) Circular	(P) $x + y = \frac{1}{2}$
(II) $a = \frac{1}{2}$	(ii) Hyperbolic	(Q) $x + y = 1$
(III) $a = 1$	(iii) Linear	(R) $x + y = \sqrt{2}$
(IV) $a = 2$	(iv) Parabolic	(S) $x + y = 4$

8. Which of the following options is the only **CORRECT** combination?
 (A) (III) (iii) (P) (B) (IV) (i) (S) (C) (II) (iv) (P) (D) (I) (ii) (R)
9. Which of the following options is the only **INCORRECT** combination?
 (A) (III) (iii) (Q) (B) (I) (ii) (S) (C) (III) (iii) (P) (D) (II) (iv) (P)
10. Which of the following options is the only **CORRECT** combination?
 (A) (II) (iii) (P) (B) (I) (ii) (R) (C) (IV) (i) (Q) (D) (IV) (i) (R)

Space for Rough Work

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given the three columns of the following table.

Match the various functions $f(x)$ given in **Column-I** with their **RANGE** given in **Column-II** and a property of the real number α such that $f(\alpha) = 1$, given in **Column-III**.

($\{\cdot\}$ and $[\cdot]$ respectively denote the fraction part and the greatest integer function)

Column-I (Function)	Column-II (Range)	Column-III ($f(\alpha) = 1$)
(I) $f(x) = [x] \{x\}$	(i) $(-\infty, \infty)$	(P) $\alpha_{\max.} = 1$
(II) $f(x) = [x] \{x\}$	(ii) $[0, 2)$	(Q) $ \alpha _{\min.} = \frac{3}{2}$
(III) $f(x) = \left[x \left[\frac{1}{x} \right] \right]$	(iii) $[0, \infty)$	(R) $ \alpha _{\min.} = \frac{5}{2}$
(IV) $f(x) = \{x\} + \left\{ \frac{1}{x} \right\}$	(iv) $\{k : k \in \mathbb{I}, k \geq 0\}$	(S) $ \alpha - 1 _{\min.} = \frac{\sqrt{5} - 1}{2}$

11. Which of the following options is the only **CORRECT** combination?
 (A) (I) (iii) (Q) (B) (I) (i) (R) (C) (III) (iv) (R) (D) (II) (iii) (S)
12. Which of the following options is the only **CORRECT** combination?
 (A) (I) (i) (P) (B) (IV) (iii) (S) (C) (III) (ii) (Q) (D) (II) (iii) (Q)
13. Which of the following options is the only **INCORRECT** combination
 (A) (IV) (ii) (S) (B) (III) (iv) (P) (C) (III) (iv) (S) (D) (I) (i) (R)

Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)
 No question will be asked in section II

SECTION-III : (Maximum Marks : 15)

- This section contains **FIVE** questions.
 - The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
 - For each question, darken the bubble corresponding to the correct integer in the ORS.
 - For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.
-

1. Let $\tan^{-1} x = \frac{1}{2} \tan^{-1} y = \frac{1}{3} \tan^{-1} z = \frac{\pi}{7}$, then the value of $\frac{1}{2}(x^2 + y^2 + z^2 - x^2 y^2 z^2)$, is
 2. Let for the function $f(x) = 5a^2 x^3 + 6ax^2 - 27x + 3b$, ($a \in \mathbb{R}^-$, $b \in \mathbb{R}$), $f'\left(-\frac{5}{9}\right) = 0$, $f''\left(-\frac{5}{9}\right) < 0$ and there exists some $\alpha \neq -\frac{5}{9}$ such that $f'(\alpha) = 0$, $f(\alpha) > 0$. The minimum possible integral value of $(a + b)$, is
-

Space for Rough Work

3. Let $I = \int_{-3}^{-2} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 \left(1 + \frac{1}{x^2} + \frac{1}{(1-x)^2} \right) dx$ then the greatest integer less than the value of I, is
4. Let f be a function such that $xf(x) = \pi \sin x \forall x \in (0, \pi)$ and $f(x) + f(\pi - x) = g(x)$ then $g_{\max}$ is

Space for Rough Work

5. Let $f(x) = (x - 1) |\operatorname{sgn}(x)|$ and $g(x) = \max.(f(t)), -\infty < t \leq x, -\infty < x < \infty$, then the number of points where $g(x)$ is non-differentiable is
($\operatorname{sgn}(x)$ represents signum function of x)

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	7	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	28
I(ii)	Matching Lists Type (Single correct option)	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18
III	Single digit Integer (0-9)	5	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	15

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator